

DAMI

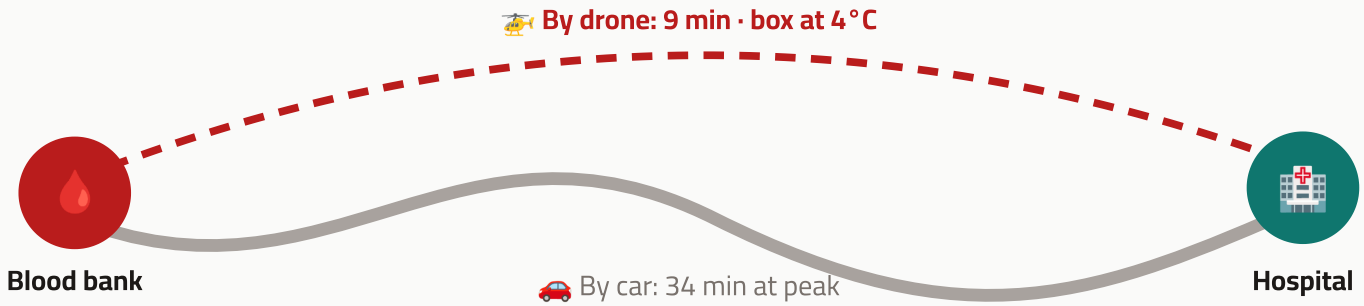
The right blood unit, to the right hospital, before it's requested… and by drone when there's no time

Because blood that arrives late… doesn't arrive.

5 p.m. — an accident on the Airport Road

A hospital needs **6 units of O negative** now. The car from the blood bank needs 34 minutes in traffic.

In severe bleeding, every minute matters.



The problem: shortage here... waste there

Blood banks face a hard equation: shortages of certain blood types when they are needed, while units expire elsewhere. In major accidents or shortage crises, hospitals need urgent units from the blood bank or another hospital, but vehicles get stuck in traffic or delayed on mountain and remote roads, and in severe bleeding every minute counts.

35–42 days

Shelf life of red blood cells stored at 2–6°C

5–7 days

Shelf life of platelets only, so some are wasted unless distributed smartly

O negative

The type requested first in severe bleeding, before the patient's type is known

Since 2016

Drone blood delivery operates in Rwanda, then Ghana and other countries (Zipline)

The solution: DAMI — three layers, one platform



Forecasting & balancing

A model forecasts 14-day demand for each type at each hospital, warns before shortages, and proposes moving units close to expiry to a hospital that will use them.



Smart donor call-up

Selects eligible donors (last donation 90+ days ago) who are nearby and most likely to respond, and invites them for exactly the type that's short.



Blood ambulance drone

In emergencies the system picks the nearest source holding the type, compares drone vs car time, and launches a drone with a smart cooling box and live temperature and position tracking.

A Jordanian AI platform that forecasts hospitals' blood needs, rebalances stock before it expires, calls the right donors, and in emergencies launches a cooled medical drone that carries blood from the blood bank to the hospital in minutes.

The drone mission: blood bank to hospital in minutes

1. Emergency request
The hospital requests 6 units of O negative in the app with “critical” priority.

2. Source selection
The system checks live stock and picks the blood bank or the nearest hospital holding the type.

3. Launch decision
Comparison: car 34 min at peak vs drone 9 min; the operator approves with one tap.

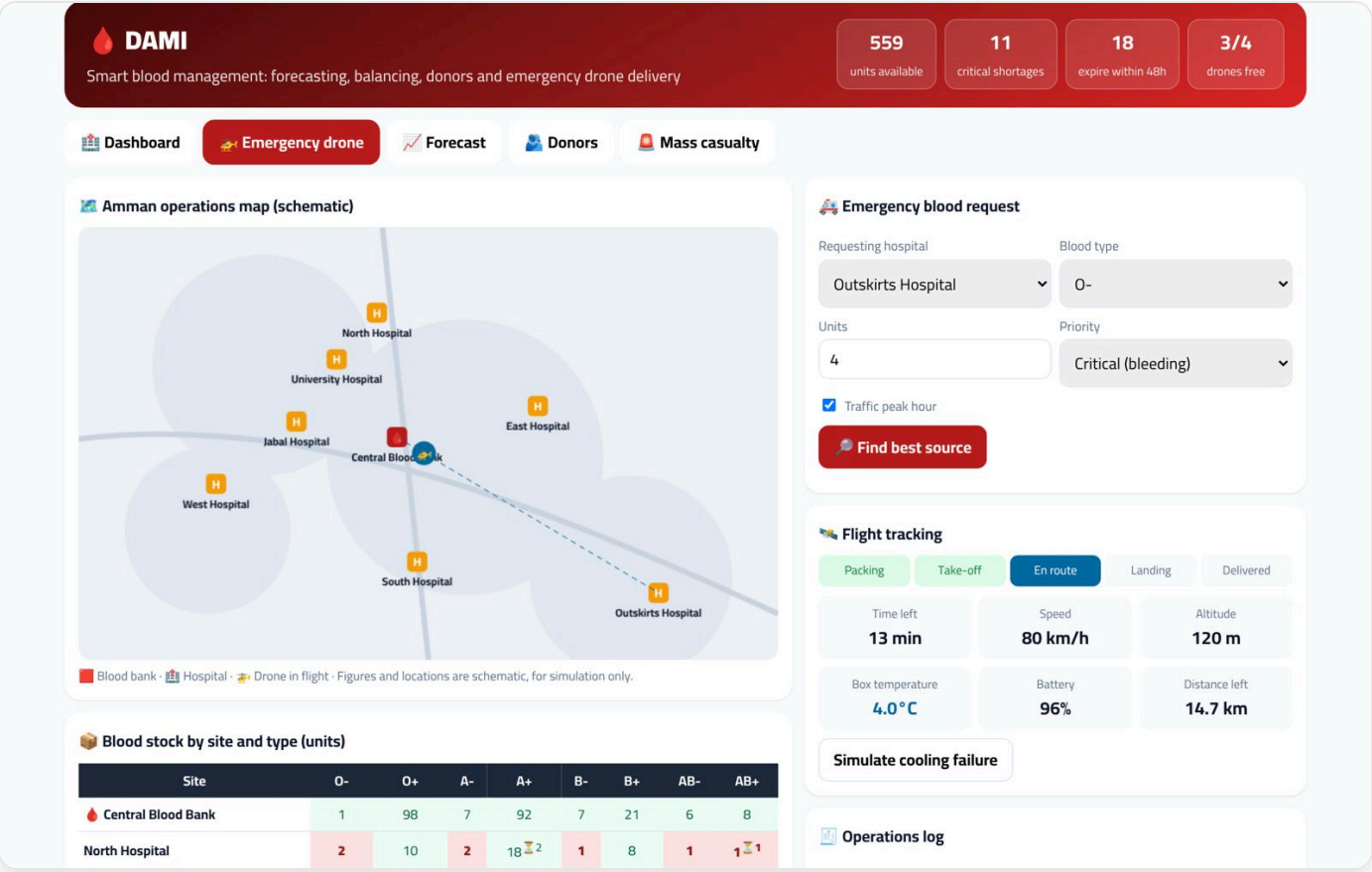
4. Packing & flight
Units packed in a 2–6°C cooled box, on a safe route that avoids restricted zones.

5. Delivery & restock
Landing on the hospital pad, receipt and flight temperature confirmed, and donors called to restock.

Type	Multirotor drone for the pilot, then fixed-wing VTOL for longer distances
Payload	2–4 kg: 4–6 blood units with the cooled box
Range & speed	15–40 km for urban pilots, at 60–90 km/h
Cooling box	Thermal insulation and phase-change material holding 2–6°C, with a temperature logger and live alarm

Connectivity	4G/5G cellular with backup link, streaming position, temperature and battery every second
Safety	Emergency parachute, automatic return on link loss, and geofencing
Delivery	Landing on a dedicated pad on the hospital roof or a small-parachute drop in a safe zone
Operations	A licensed pilot monitors from the operations center, on pre-approved routes

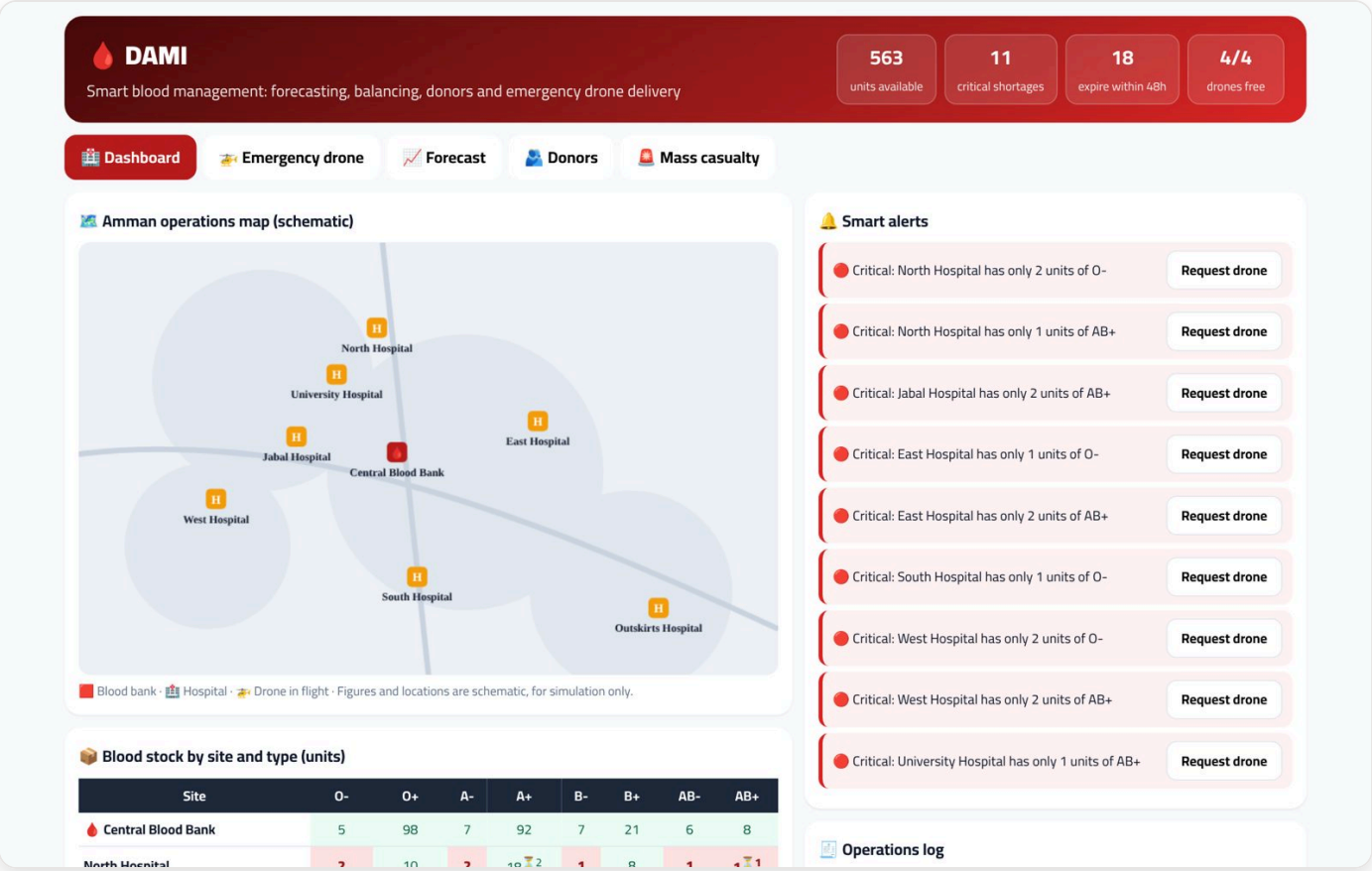
The prototype works: Emergency drone request



Best-source selection, drone vs car time comparison, then launch and tracking: temperature, battery, altitude and distance.

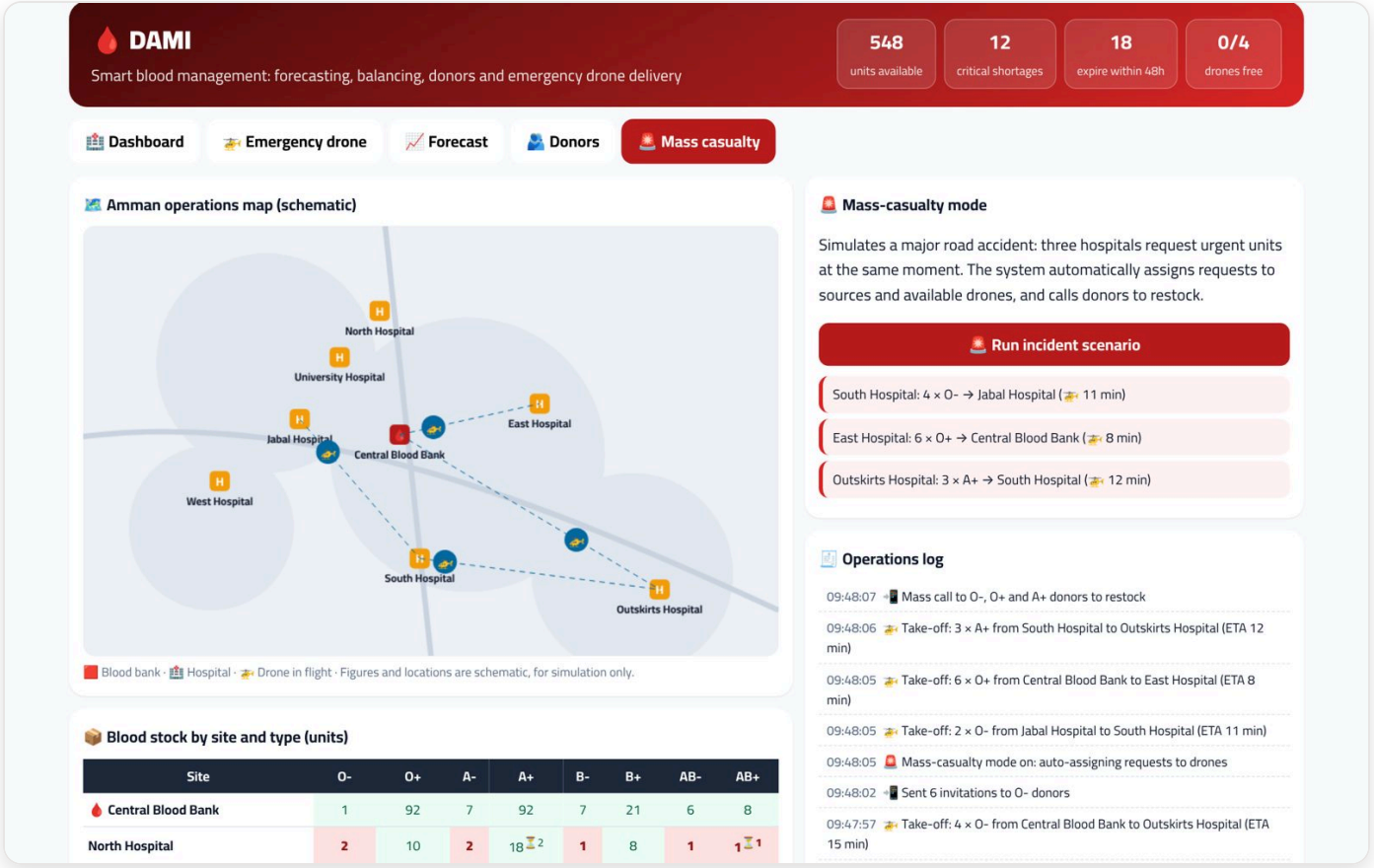
⚡ A simulated accident on a main road: a hospital requests 6 units of O negative immediately. Within seconds the system picks the blood bank and shows the comparison: car 34 minutes vs drone 9 minutes. The drone takes off on the map, the box holds steady at 4 °C, and eligible donors receive messages to restock.

The prototype works: Dashboard



Stock of 8 blood types at the blood bank and 7 hospitals, color-coded shortage and expiry alerts, and one-tap transfer or drone request.

The prototype works: Mass-casualty mode



3 hospitals request blood at the same moment; the system dispatches drones automatically and calls donors.

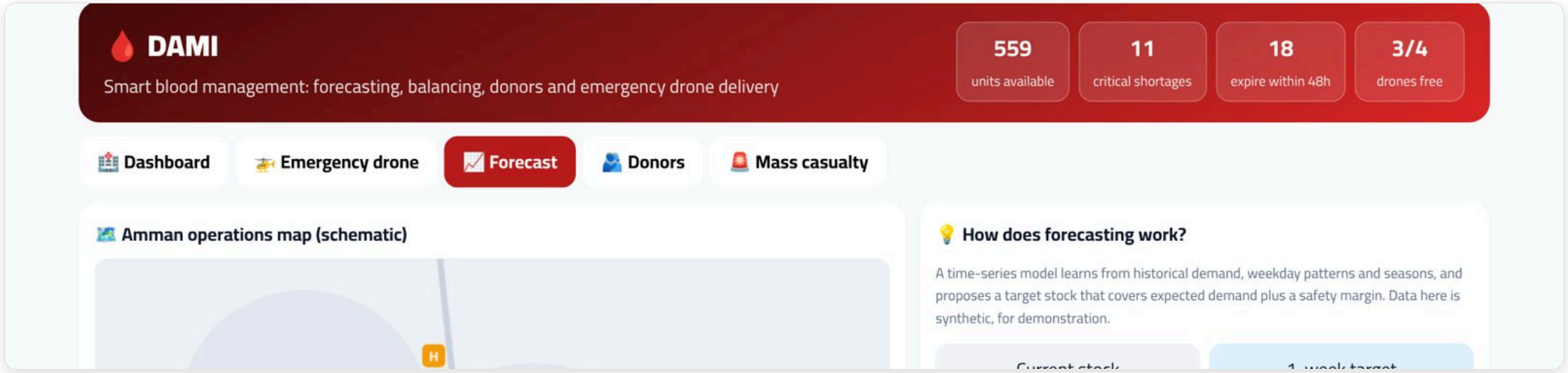
The AI inside DAMI

Models

- Time-series forecasting of demand per blood type
- Optimization of stock distribution and best-source selection
- Drone route planning around weather and restricted zones
- Selecting the donors most likely to respond

Data

- Blood bank inventories
- Request history
- Donor database (with consent)
- Weather and airspace data
- Temperature sensor readings from transport boxes



Market and competition

Alternative	What it offers	Where it falls short
Status quo	Phone calls between blood banks, transport by car, general donation drives	Slow in emergencies, expiry waste, no type targeting
Blood bank management systems	Stock records inside one hospital	No forecasting, no national view, no delivery
Zipline (global)	Drone delivery in Africa and the US	Not present in Jordan per our preliminary research; focused on delivery only
DAMI	**Forecasting + balancing + donors + drone in one platform**	**Built for Jordan; starts with simulation and phases into licensed operations**

We start with private hospitals because they decide faster, then the Ministry of Health and the National Blood Bank.

Business Model Canvas

Key Partnerships

- Ministry of Health & National Blood Bank
- Private and public hospitals, Royal Medical Services
- Medical drone operators and manufacturers
- Civil Aviation Regulatory Commission & security authorities
- Jordan Red Crescent
- Telecom and insurance companies, CSR sponsors

Key Activities

- Building forecasting and balancing models
- Running the emergency ops center and fleet
- Cold-chain and quality management
- Running the donor app and call-up campaigns
- Building partnerships and obtaining permits

Key Resources

- Software platform and AI models
- Drone fleet and smart cooling boxes
- Consented donor database
- Tech team, licensed pilots and a blood bank expert
- Permits and agreements

Value Propositions

- Hospitals: emergency units in minutes instead of half an hour
- Blood banks: less waste by rebalancing before expiry
- Ministry: a live national dashboard and mass-casualty readiness
- Donors: invited at the right time and see their impact
- Patients: blood available when they need it

Customer Relationships

- Annual service contract with an account manager
- 24/7 emergency operations center
- Monthly impact reports
- Donor community, rewards and badges

Channels

- Direct sales to hospital management
- Partnership with the Ministry of Health and the blood bank
- Health conferences and exhibitions
- Media campaigns and universities for donors
- Mobile app and web dashboard

Customer Segments

- Private hospitals (beachhead)
- Ministry of Health hospitals and Royal Medical Services
- National Blood Bank
- Red Crescent and Civil Defense
- Donors (free users)

Cost Structure

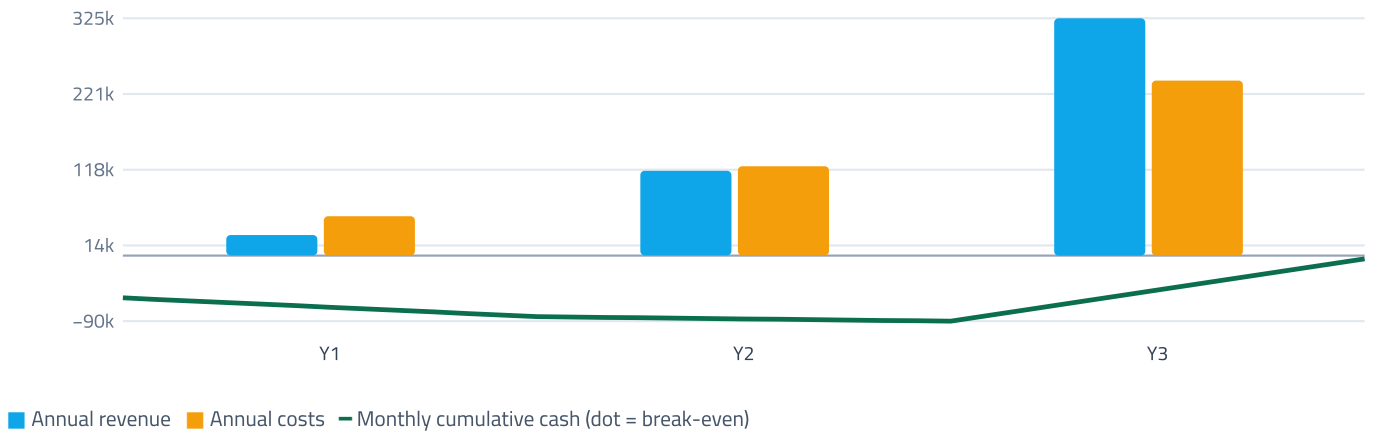
- Software and AI development
- Drones and cooling boxes, purchase and maintenance
- Insurance, permits and compliance
- Team and pilot salaries
- Servers, connectivity and marketing

Revenue Streams

- Annual hospital subscription: JOD 6,000
- Emergency drone flight fee: JOD 90
- National dashboard contract: JOD 40,000/year
- Corporate sponsorship of the donor app: JOD 10,000/year

The numbers: how we make money

Stream	Unit	Price	Y1	Y2	Y3	Y3 revenue
Hospital subscription	hospital/year	JOD 6,000	3	10	20	JOD 120,000
Emergency drone delivery	flight	JOD 90	0	400	1,500	JOD 135,000
National dashboard contract	annual contract	JOD 40,000	0	0	1	JOD 40,000
Donor-app sponsorship	sponsor/year	JOD 10,000	1	2	3	JOD 30,000



Initial investment
JOD 58,000

Year-3 revenue
JOD 325,000

Year-3 gross margin
72%

Year-3 operating result
JOD 85,250

Break-even (optimistic)
Month 27

Margin per drone flight
55%

The plan: from simulator to first licensed flight

Phase 1 · Months 0–6	Prototype and full simulator (ready) + synthetic data + interviews with 10 blood banks and hospitals.
Phase 2 · Months 6–12	Software pilot in 3 private hospitals at a discounted subscription: forecasting, balancing and donors.
Phase 3 · Months 12–24	Pilot drone route between a blood bank and a hospital after flight permits, with paid flights.
Phase 4 · Months 24–36	National dashboard under a government contract, expanding the fleet to remote governorates and mass-casualty response.

Risks and mitigation

Risk	Mitigation
Sensitivity of health data	Formal partnership, encryption and consent
Licensing drone flights in Jordanian airspace	Start with simulation, then a pilot route approved by the Civil Aviation Regulatory Commission and security authorities
Maintaining the cold chain	Medically certified boxes with live temperature monitoring and alarms
Slow government contracting	Start with private hospitals and corporate sponsorship

Impact metrics

- Units saved from expiry
- Minutes saved in emergency delivery
- Share of shipments within safe temperature
- Number of active donors

Team and the ask

Project Lead / AI Lead	Technical architecture, forecasting models and coordination
Data Scientist	Stock and demand data, and model evaluation
Systems Engineer / Drones & IoT	Drone, cooling box and sensor integration
App Developer	Dashboard, donor app and live demo
Business & Pitch Lead	Health and regulatory partnerships, business model and pitch

The Ask

- Seed funding of about JOD 89,900 covering initial investment and the growth period; the pilot fleet (JOD 30,000) can be split out as a health grant.
- A pilot partnership with a blood bank and a private hospital to connect real stock.
- Support on the permit path with the Civil Aviation Regulatory Commission and security authorities.
- Mentoring in the health sector and entrepreneurship.

Because blood that arrives late… doesn't arrive.

🩸 **DAMI · Team NEXORA · Thank you**